Chapter 4, Problem 5RQ

Problem

The Thevenin voltage across terminals *a* and *b* of the circuit in Fig. 4.67 is:

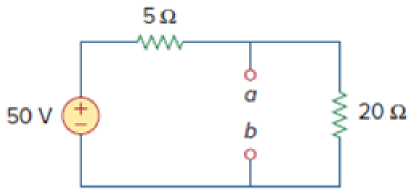
(a) 50 V

(b) 40 V

(c) 20 V

(d) 10 V

Figure. 4.67:



Step-by-step solution

Step 1 of 2

Refer to the circuit diagram in Figure 4.67 in the textbook.

The Thevenin voltage at terminals *a* and *b* is same as voltage across $20\ \Omega$ resistor. Since terminals *a* – *b* and $20\ \Omega$ resistor are in parallel.

Identify voltage v_T at resistor $20\ \Omega$.

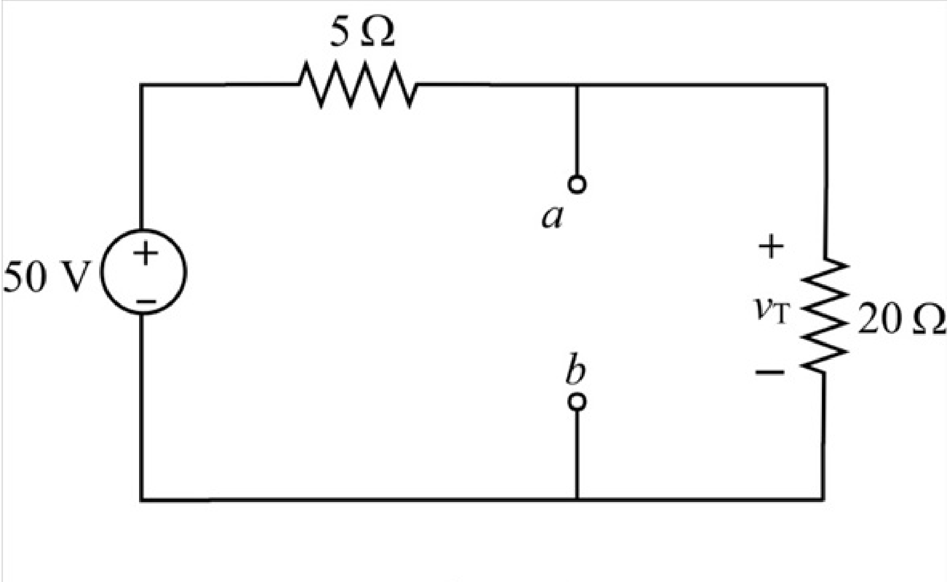


Figure 1

Step 2 of 2

From Figure 1, find the thevenin voltage at terminals *a*-*b* using voltage division rule.

$$\begin{aligned} v_T &= 50\left(\frac{20}{20+5}\right) \\ &= 50\left(\frac{20}{25}\right) \\ &= 50\left(\frac{4}{5}\right) \\ &= 40\ \text{V} \end{aligned}$$

Thus, thevenin voltage v_T at the terminals *a* and *b* is $40\ \text{V}$.

So correct option is $\boxed{\text{b}}$.